PROFESSIONAL INITIATION INTERNSHIP · DEFENSE

Calibration of AI Weather Models Using Machine Learning

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2-month internship (remote) · Academic year 2025/2026

One question, answered in three steps

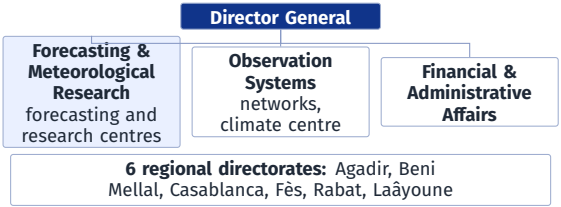
1. Context: DGM, and why AI forecasts need a local correction
2. **The question** I tried to answer
3. Data, then **measure and explain** the error (statistics)
4. **Correct** the error (machine learning: the mathematics)
5. Results, limits, and the **answer**
6. **What I learned**

Thread of the talk

question → measure → explain → correct → verify → answer

DGM must know how AI forecasts behave at its own stations

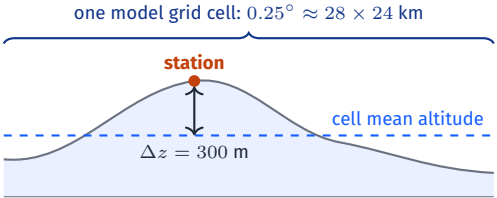
- Morocco's public weather service since **1961** (*Maroc Météo*), under the Ministry of Equipment and Water
- Observes the atmosphere and the sea, keeps the climate archive, issues forecasts: **protects people and property**
- Serves aviation, agriculture, maritime transport, water management, civil protection
- ISO 9001 since 2014; 6 regional directorates; its research includes **machine learning**



Simplified from the DGM organisation chart (report, Fig. 1.1)

Why this subject? Developers score AI models on global grids, not at Moroccan stations: before using them, DGM needs a **local** evaluation.

A grid cell is not a station, so part of the forecast error is systematic



Altitude example (report, §2.3)

$\Delta T \approx \Gamma\, \Delta z = 6.5 \, \frac{°\mathrm{C}}{\mathrm{km}} \times 0.3\,\mathrm{km} \approx 2\,°\mathrm{C}$
$\Rightarrow$ the model reads about 2 °C too warm, **every day, in the same direction**.

- **AI weather models** (since 2022) learn from decades of past data (ERA5) and forecast in minutes. AIFS is operational at ECMWF since 25 Feb 2025.
- The model gives an **average over a cell**; the station measures **one point**, 2 m above ground: a gap is expected.
- Learned models also produce **smooth fields** that miss extremes.

bias = predicted value − observed value
A bias that repeats is **systematic**: it can be learned from past pairs and subtracted (post-processing, MOS since 1972).

The question I tried to answer

> **How large is the error of AI weather models at Moroccan stations, what explains it, and how much of it can machine learning remove?**

Q1 · Measure

Error of **8** AI models at **34** stations, for **5** weather variables.

Q2 · Explain

Does it depend on the **hour**, the **forecast horizon**, the **station**?

Q3 · Correct and verify

Learn the error, subtract it, and check on **later, unseen dates** by how much it improves.

Core difficulty: find the inputs the correction needs, and *quantify* the error it removes.

Short answer: t2m RMSE reduced for all 8 models (−14.7 % to −44.7 %); precipitation not yet.

261,082 forecast–observation pairs: 8 AI models, 34 stations, 5 variables

Period	Sep 2025 – Mar 2026 (7 months)
Valid times	00, 06, 12, 18 UTC
Lead time	0 to 72 h
Stations	34: 17 coastal, 13 inland, 4 mountain or pre-Saharan
Grid distance	2.4 to 65 km
Missing values	none

AI model	Developer
AIFS	ECMWF
Aurora	Microsoft
DLWP	University of Washington
FengWu	Shanghai AI Laboratory
FourCastNet	NVIDIA
FuXi	Fudan University
GraphCast	Google DeepMind
Pangu6	Huawei

One row = one forecast and the observation at the same moment:

$$\text{valid time} = \text{run time} + \text{lead time}$$

Variables: t2m (°C), td2m (°C), msl (hPa), sp (hPa), tp06 (mm). **t2m** exists for every model (21,474 to 24,611 pairs each): main target. tp06 (3 models): second target.

The mean bias alone hides the error: AIFS is centred on zero but has the largest RMSE

Three scores, with $b_i = \hat{y}_i - y_i$

$$\bar{b} = \frac{1}{n}\sum_i b_i$$

$$\mathrm{MAE} = \frac{1}{n}\sum_i |b_i|$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i b_i^2}$$

$\hat{y}_i$: AI forecast, y_i: observation

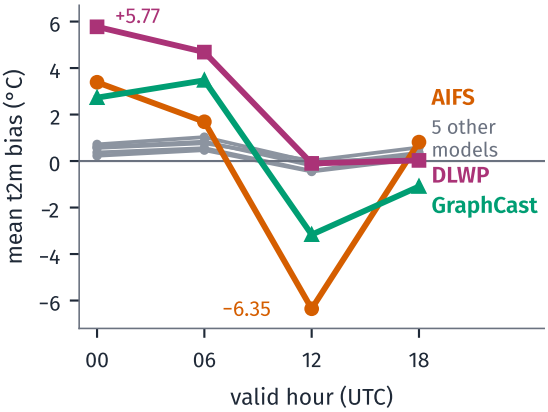
- In $\bar{b}$, positive and negative errors **cancel**.
- RMSE squares errors: one 10 °C error costs more than ten 1 °C errors.

t2m, raw forecast	$\bar{b}$ (°C)	MAE (°C)	RMSE (°C)
Aurora	+0.18	1.61	2.12
FengWu	+0.09	1.64	2.14
Pangu6	+0.47	1.67	2.19
FuXi	+0.48	1.71	2.23
FourCastNet	+0.40	1.80	2.34
GraphCast	+0.47	3.55	4.51
DLWP	**+2.57**	3.94	5.08
AIFS	**−0.15**	3.94	**5.13**

All stations and lead times, 21,474 pairs per model (AIFS 24,611); report Table 4.1

Two groups: five models at **2.1–2.35 °C**, three at **4.5–5.1 °C**.

For AIFS, DLWP and GraphCast, the error follows the hour of the day



Mean t2m bias per valid hour (report Fig. 4.3)

One-way ANOVA over the 4 valid hours

$$H_0: \ \mu_{00} = \mu_{06} = \mu_{12} = \mu_{18}, \quad k = 4$$

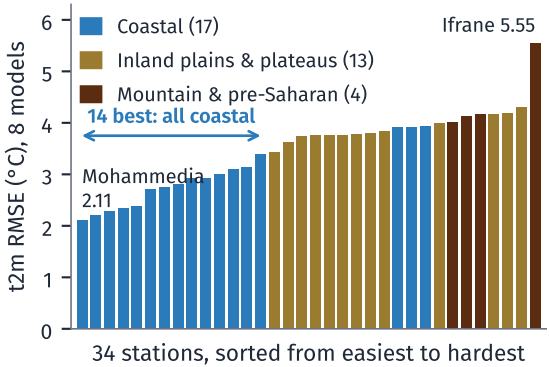
$$F = \frac{\sum_g n_g(\bar{b}_g - \bar{b})^2/(k-1)}{\sum_g \sum_{i \in g}(b_i - \bar{b}_g)^2/(n-k)}$$

AIFS	9072	GraphCast	4212
DLWP	4160	5 others	121–297

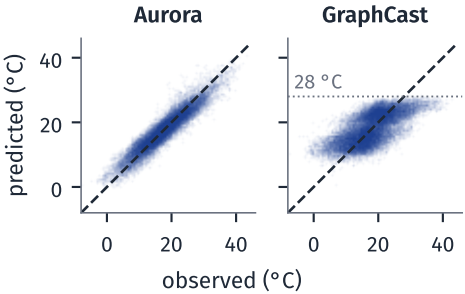
With $n > 21{,}000$ per model, $p \approx 0$ for *any* difference: the information is in the **size of** F.

Physics: coarse models over-mix the air near the ground, so noon is forecast too cold and night too warm.

Station location and temperature range also shape the error



Coastal cells are flat and regulated by the sea; the Ifrane cell averages plateaus and valleys.



All 8 models predict **cold days too warm and hot days too cold**; best between 10 and 25 °C. GraphCast almost never predicts above 28 °C; observations reach **42.7 °C**.

Recomputed from the dataset, same values as report Fig. 4.7 (groups of Table 3.5) and Fig. 4.6b.

Mutual information separates the eight models into two families

What feature X tells about the bias Y

$$I(X;Y) = \iint p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)}\, dx\, dy$$

$I = 0 \iff X, Y$ independent. No linearity assumed: Pearson's correlation is near 0 for the curved hour–bias link.

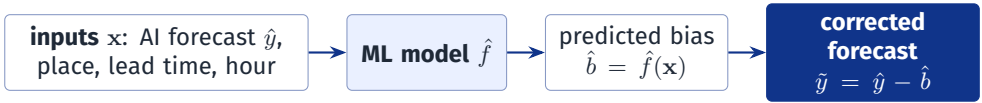
- **Time-driven**: hour, lead time 0.25–0.37
- **Location-driven**: station, lat, lon 0.12–0.22; hour < 0.03

	hour	lead time	station	lat	lon	month	
AIFS	0.36	0.37	0.14	0.05	0.05	0.02	time-driven
DLWP	0.25	0.27	0.14	0.11	0.11	0.06	
GraphCast	0.25	0.25	0.16	0.07	0.07	0.04	
Aurora	0.03	0.03	0.19	0.13	0.13	0.01	location-driven
FengWu	0.02	0.02	0.20	0.14	0.15	0.02	
FourCastNet	0.02	0.02	0.16	0.12	0.12	0.01	
FuXi	0.01	0.01	0.22	0.16	0.16	0.01	
Pangu6	0.03	0.03	0.15	0.12	0.12	0.01	

Mutual information with the t2m bias (report Fig. 4.10)

⇒ The correction must know **when** and **where** the forecast is valid.

The correction learns the bias from what is known at forecast time



Supervised learning problem

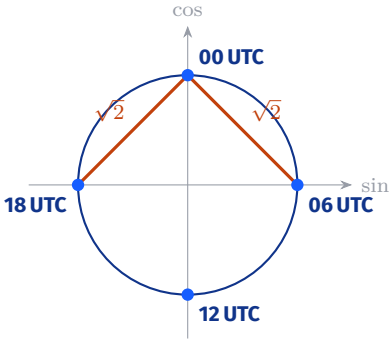
Data $(\mathbf{x}_i, b_i)_{i=1}^n$ with $b_i = \hat{y}_i - y_i$. Find f with $f(\mathbf{x}) \approx b$ on data **not seen** in training, e.g. by minimising

$$\mathcal{L}(f) = \frac{1}{n} \sum_{i=1}^{n} \big(b_i - f(\mathbf{x}_i)\big)^2.$$

Rule: the observation y is **never** an input. It would make the bias almost perfectly predictable, but at forecast time y does not exist.

- **Config A** (8 inputs): $\hat{y}$, lat, lon, lead time, $\sin$, $\cos$ of the hour, is_night, is_afternoon
- **Config B**: A + the other variables predicted by the same AI model

The hour of the day enters the model as a point on the unit circle



Problem: as numbers, hours 18 and 00 are 18 units apart, yet only 6 h apart in time.

Encoding of the valid hour h

$$h \longmapsto \left(\sin \tfrac{2\pi h}{24},\ \cos \tfrac{2\pi h}{24}\right) \in S^1$$

18 h→00 h and 00 h→06 h are now at the same distance $\sqrt{2}$. Both coordinates are needed: $\sin 0 = \sin \pi = 0$, so the sine alone confuses 00 and 12 UTC.

Two binary features, chosen from the data: is_night $= 1$ at 00 and 06 UTC, is_afternoon $= 1$ at 12 UTC. A tree needs one split on is_night, several on $(\sin, \cos)$.

Four algorithms from four families, all fitting $f(\mathbf{x}) \approx b$

Multi-layer perceptron

$$\mathbf{a}^{(l)} = \tanh\big(W^{(l)}\mathbf{a}^{(l-1)} + \mathbf{c}^{(l)}\big), \quad \mathbf{a}^{(0)} = \mathbf{x}$$

Layers 64 and 32; MSE loss; Adam; early stopping. Inputs standardised.

Support vector regression, RBF kernel

$$f(\mathbf{x}) = \sum_{i=1}^{n} (\alpha_i - \alpha_i^*)\, K(\mathbf{x}_i, \mathbf{x}) + b_0$$

Kernel $K(\mathbf{u}, \mathbf{v}) = e^{-\gamma \|\mathbf{u}-\mathbf{v}\|^2}$. Errors inside a tube of width ε cost nothing.

Random Forest: independent trees

$$f(\mathbf{x}) = \frac{1}{B} \sum_{j=1}^{B} T_j(\mathbf{x})$$

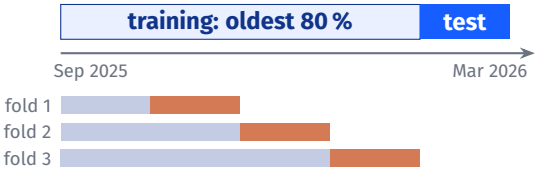
Bootstrap rows, random features; averaging lowers variance.

LightGBM: sequential boosted trees

$$f_K(\mathbf{x}) = \sum_{k=1}^{K} \eta\, T_k(\mathbf{x})$$

Penalty $\Omega(T) = \gamma L + \frac{1}{2}\lambda \sum_j w_j^2$ (L leaves, values w_j).

Validation respects time: train on the past, test on the most recent 20 %



- Sorted by valid time, split per model and variable, **never shuffled**: a random split would train on days after the test days.
- t2m, Config A: AIFS 19,688 / 4,923 rows; others 17,179 / 4,295. Scaler fitted on training rows only.

Level 1: the predicted bias

MAE, RMSE and

$$R^2 = 1 - \frac{\sum_i (b_i - \hat{b}_i)^2}{\sum_i (b_i - \bar{b})^2}$$

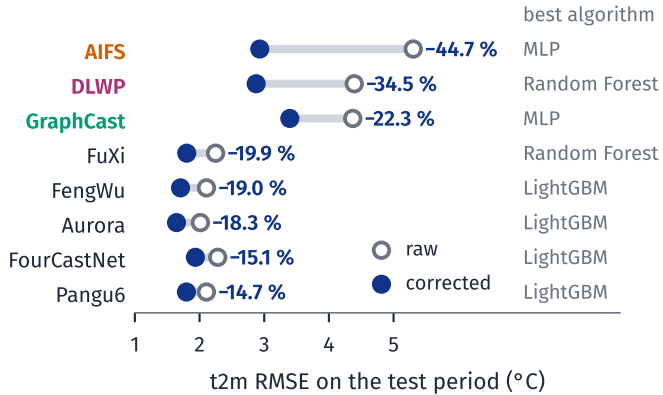
$R^2 < 0$: worse than predicting the mean bias.

Level 2: the forecast (used to compare)

$$\text{Gain} = \left(1 - \frac{\text{RMSE}_{\text{corrected}}}{\text{RMSE}_{\text{raw}}}\right) \times 100\,\%$$

Same test rows, same observations.

Correction lowers the t2m RMSE of all eight models, by 14.7 to 44.7 %



Config A, best of 4 algorithms, most recent 20 % of dates (report Table 4.2)

- The best algorithm **follows the family**: MLP for time-driven, trees for location-driven.
- Biggest raw errors, biggest gains; yet AIFS stays at **2.93 °C**.
- SVR always behind: errors inside its tuned ε-tube ($\varepsilon = 0.483$, standardised target) are not penalised.

Config B: + other variables

AIFS **−49.0 %** ($R^2 = 0.739$), but on 842 train / 211 test rows: *promising, not established*.

Precipitation: rain errors are detected, but with about 30 false alarms per real case

98.48 % of tp06 observations are 0 mm; only 1.3–2.0 % of rows have $|b| > 0.1$ mm. A plain regression learns "bias $= 0$".

Two-step method

1. **Classify:** is $|b| > 0.1$ mm? Rare class weighted by $n_{\text{neg}}/n_{\text{pos}}$; test set untouched.
2. **Regress** b on the significant cases.

Recall (POD) **0.50–0.88**, precision $\leq$**0.032**: at least 30 false alarms per real case. Step 2: R^2 0.17–0.22 at best; 9 of 20 runs $R^2 < 0$.

	predicted yes	**predicted no**
observed yes	TP (hit)	FN (miss)
observed no	FP (false alarm)	TN

Scores from the counts

$$\text{Recall} = \frac{TP}{TP+FN} = \text{POD}$$

$$\text{Precision} = \frac{TP}{TP+FP} = 1 - \text{FAR}$$

$$F_1 = \frac{2\,\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}$$

Always raising an alarm: POD $= 1$ but FAR ≈ 0.99. Read both.

What the results do not show yet, and what comes next

Limits

- 7 months, no summer; test period: late winter and early spring
- 17 coastal stations, only 4 mountain or pre-Saharan, and those are the hardest
- Config B on small samples (AIFS: 211 test rows)
- Precipitation: not a working method yet
- Not tested at stations unseen in training

Next steps

- One to two full years of data: summer, usable Config B
- Leave-one-station-out test
- Learned station embeddings instead of lat/lon
- Probabilistic output (mean and spread, CRPS)
- Rain: tune the threshold, add humidity
- Operational chain with monthly retraining

CONCLUSION

Answer: the error is systematic, structured, and largely removable for temperature

Q1 Measure Five models: t2m RMSE **2.1–2.35 °C**; AIFS, DLWP, GraphCast: **4.5–5.1 °C**. All eight compress the temperature range.

Q2 Explain The **hour** for AIFS, DLWP, GraphCast (AIFS: +3.4 °C at night, −6.4 °C at noon); the **station** for the five others. ANOVA, mutual information and Random Forest importance agree.

Q3 Correct t2m RMSE **−14.7 to −44.7 %** on later, unseen dates; up to −49.0 % for AIFS with more inputs (small sample). Rain: not corrected in practice.

Main lesson: the algorithm must match the structure of the error. A neural network wins when the bias is a nonlinear function of time; trees win when it depends on the place. This split appears only because eight models were compared together.

What I learned: from course formulas to an open, real problem

Mathematics, statistics

- A mean can hide everything (AIFS: $\bar{b} = -0.15\,°\text{C}$, RMSE 5.13 °C).
- With $n > 21{,}000$, $p \approx 0$: read the effect size.
- Flexibility must match data size: tuning on 842 rows did worse.

Engineering method

- Explore first: the analysis chose the features and algorithms.
- No leakage: split by time; the observation is never an input.
- Compare with the raw forecast on the same rows, not by R^2 alone.

Professional

- Remote work through my supervisor: precise questions, freedom to test my own ideas.
- An open problem, no predefined solution.
- A service validates a tool locally before using it.

Difficulty → **action** → **result:** rain is 98 % zeros → two-step method, class weights → events found, too many false alarms.

Thank you for your attention

1. AI forecasts carry a **systematic** error at Moroccan stations: 2.1 to 5.1 °C RMSE for t2m.
2. It is driven by the **hour** for three models and by the **station** for five.
3. A learned correction removes **14.7 to 44.7 %** of the t2m RMSE; rain remains open.

Thanks to Dr. ALAOUI Badreddine, to the DGM, and to the EHTP administration and teachers.

Questions?

Backup: all four algorithms on temperature, Config A

RMSE gain, test period	MLP	SVR	Random Forest	LightGBM	raw RMSE (°C)
AIFS	**44.7 %**	43.3 %	44.5 %	43.6 %	5.30
DLWP	34.3 %	31.9 %	**34.5 %**	33.7 %	4.39
GraphCast	**22.3 %**	19.2 %	21.6 %	20.9 %	4.37
FuXi	19.1 %	8.2 %	**19.9 %**	19.7 %	2.25
FengWu	17.0 %	6.4 %	19.0 %	**19.0 %**	2.11
Aurora	17.2 %	6.2 %	17.7 %	**18.3 %**	2.01
FourCastNet	13.5 %	5.7 %	14.6 %	**15.1 %**	2.28
Pangu6	11.3 %	5.2 %	14.4 %	**14.7 %**	2.11

Report Table A.1; best per row in bold (FengWu: Random Forest 1.707 vs LightGBM 1.706 °C).

Test rows: AIFS 4,923, other models 4,295. R^2 of the best bias model: AIFS 0.694, DLWP 0.542, others 0.258 to 0.344.

Backup: Config B, adding the other predicted variables

Model	Best algorithm	R^2	Raw RMSE	Corr. RMSE	Gain	Train rows
AIFS	Random Forest	0.739	4.93	2.51	49.0 %	**842**
GraphCast	LightGBM	0.341	4.21	3.17	24.6 %	5,543
FuXi	Random Forest	0.315	2.21	1.81	18.3 %	5,705
FengWu	Random Forest	0.330	2.05	1.68	18.1 %	5,549
Aurora	LightGBM	0.292	1.96	1.63	16.7 %	5,712
FourCastNet	LightGBM	0.250	2.33	1.99	14.4 %	2,291
Pangu6	LightGBM	0.255	2.14	1.83	14.5 %	5,549

Report Table 4.3. RMSE in °C. DLWP absent: it provides temperature only.

A row is kept only if every variable of the model exists for the same station, run and lead time (gaps filled from the nearest lead time, then incomplete rows dropped). The MLP loses ground here: about 5 % gain on Aurora, FengWu, Pangu6.

Backup: hyperparameter tuning, and why a small sample misleads

Tuned on AIFS t2m (19,688 rows), applied elsewhere

RMSE change	DLWP	GraphCast	FengWu
MLP	+1.3 %	+1.1 %	+1.4 %
SVR	−1.4 %	+0.8 %	−2.8 %
Random Forest	**+11.2 %**	**+9.2 %**	**+10.4 %**
LightGBM	−0.5 %	−0.3 %	+1.0 %

Report Table 3.7; positive means tuned beats default.

- 30 / 25 / 40 / 40 trials (MLP, SVR, RF, LightGBM); score $-$MSE; 3-fold `TimeSeriesSplit`.
- Config B searched on its own (AIFS: 842 rows): worse than reusing Config A values for SVR on 5/7 models and LightGBM on 7/7; better only for RF on 4/7.

> A better search score on a small sample does not mean a better model on unseen data.

MLP (64,32), tanh; SVR $C = 0.297$, $\varepsilon = 0.483$, $\gamma = 0.0785$; RF 179 trees, depth 22; LightGBM 250 trees, depth 12, $\eta = 0.0115$ (Table A.4).

Backup: Random Forest and LightGBM disagree on the binary hour features

Random Forest: impurity importance

	AIFS	DLWP	GraphCast	Aurora	FengWu	FourCastNet	FuXi	Pangu6
pred_value	0.17	0.16	0.14	0.24	0.24	0.26	0.23	0.26
lat	0.07	0.13	0.12	0.33	0.32	0.32	0.35	0.31
lon	0.06	0.13	0.13	0.33	0.35	0.32	0.34	0.30
lead_time	0.06	0.13	0.07	0.02	0.02	0.02	0.02	0.03
hour_sin	0.05	0.08	0.08	0.01	0.01	0.02	0.01	0.03
hour_cos	0.28	0.09	0.15	0.03	0.02	0.02	0.02	0.02
is_afternoon	0.20	0.05	0.08	0.03	0.02	0.02	0.02	0.01
is_night	0.11	0.22	0.23	0.01	0.01	0.02	0.01	0.04

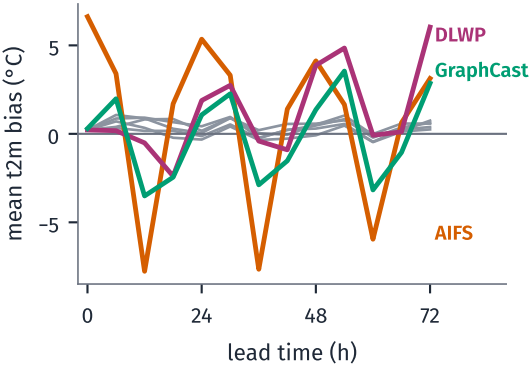
LightGBM: share of splits

	AIFS	DLWP	GraphCast	Aurora	FengWu	FourCastNet	FuXi	Pangu6
pred_value	0.33	0.28	0.41	0.31	0.35	0.33	0.34	0.33
lat	0.27	0.35	0.26	0.26	0.25	0.29	0.27	0.28
lon	0.26	0.22	0.22	0.17	0.15	0.16	0.14	0.18
lead_time	0.08	0.09	0.06	0.07	0.08	0.07	0.09	0.08
hour_sin	0.03	0.04	0.03	0.07	0.07	0.08	0.07	0.07
hour_cos	0.03	0.02	0.02	0.09	0.09	0.06	0.07	0.05
is_afternoon	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.00
is_night	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01

Config A, t2m. Same data as report Figs. 4.12 and 4.13; LightGBM split counts shown as a share per model.

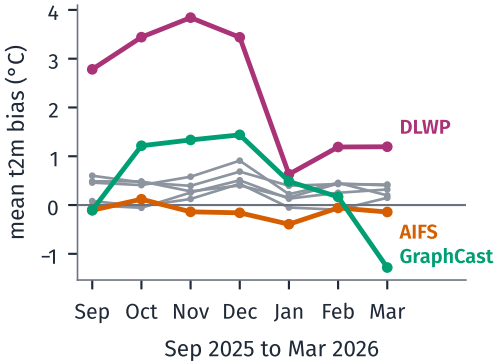
Random Forest agrees with the exploratory analysis. LightGBM counts how often a feature is *used* to split: a binary feature offers one split only, so it ranks low.

Backup: the bias against forecast horizon and month



Over 0–72 h the bias **oscillates with a 24 h period** instead of growing: the daily cycle dominates the horizon.

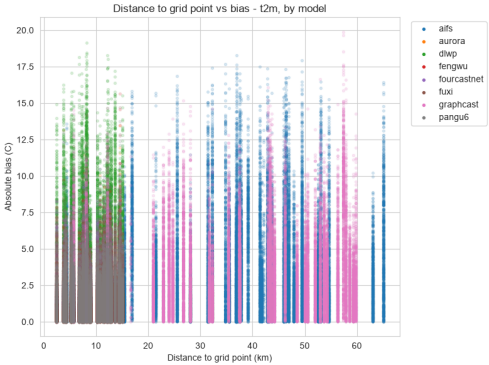
t2m panel of report Fig. 4.5, recomputed from the dataset



DLWP: +3.84 °C in Nov, +0.63 °C in Jan. GraphCast: +1.44 °C in Dec, −1.28 °C in Mar. No summer month in the record.

Report Fig. 4.8 (results_csv/bias_monthly.csv)

Backup: distance to the grid point gives a misleading correlation



Report Fig. 4.9 (as in the report)

- All models pooled: distance and absolute bias look correlated.
- Within each model: correlation close to zero.
- The pooled correlation comes from two clusters of models with different grid geometries: the architecture matters far more than the distance.

> A correlation on pooled groups can be an artefact of the grouping.

Backup: precipitation results per model and algorithm (Config A)

Model	Algorithm	Precision	Recall (POD)	F_1	R^2 (step 2)
AIFS	MLP / SVR	0.007 / 0.009	0.85 / 0.75	0.014 / 0.018	0.050 / −0.008
AIFS	RF / LightGBM	0.013 / 0.015	0.85 / 0.85	0.027 / 0.030	0.174 / 0.216
FuXi	MLP / SVR	0.021 / 0.023	0.60 / 0.70	0.040 / 0.045	−0.008 / −0.050
FuXi	RF / LightGBM	0.022 / 0.023	0.75 / 0.75	0.044 / 0.044	0.222 / 0.090
GraphCast	MLP / SVR	0.008 / 0.022	0.65 / 0.65	0.016 / 0.043	0.028 / −0.038
GraphCast	RF / LightGBM	0.023 / 0.020	0.75 / 0.60	0.044 / 0.040	0.190 / 0.066

Report Table A.3 (results_csv/calibration_results.csv). Step 2 scored on significant cases only.

Config B: 81–82 training rows and 8 test rows, too few to conclude. AIFS Config B skipped: 15 significant cases, below the minimum of 20.

Backup: the two evaluation levels are linked by an identity

From the definitions $b = \hat{y} - y$ and $\tilde{y} = \hat{y} - \hat{b}$

$$\tilde{y} - y = (\hat{y} - y) - \hat{b} = b - \hat{b}$$

So, on the same test rows:

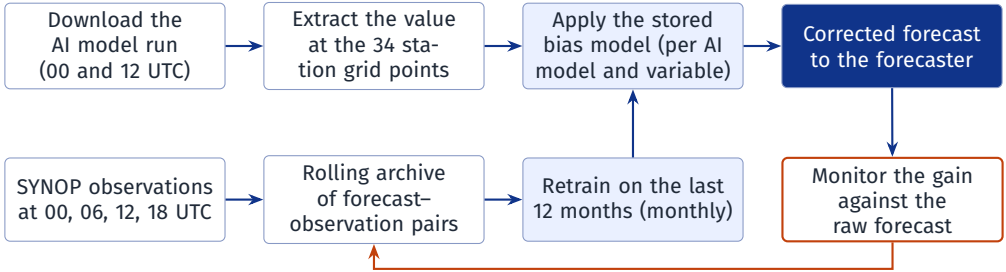
- $\mathrm{RMSE}_{\text{corrected}} = \mathrm{RMSE}(\hat{b}, b)$, the Level-1 RMSE;
- $\mathrm{RMSE}_{\text{raw}}$ is the RMSE of the trivial model $\hat{b} \equiv 0$;
- Gain $= 1 - \mathrm{RMSE}(\hat{b}, b)/\mathrm{RMSE}(0, b)$.

Check in the result file: in every experiment of `calibration_results.csv`, the Level-1 RMSE equals the corrected RMSE (AIFS, MLP: 2.931 and 2.931 °C).

AIFS, Config A: $1 - (2.931/5.304)^2 = 0.695$, next to the reported $R^2 = 0.694$: the AIFS test-period mean bias is close to 0.

Algebra on the report's definitions; no new experiment.

Backup: proposed operational chain (concept, not implemented)



Report Fig. P.1

Main risk: an AI model updated by its authors without warning can make a stored correction harmful. A drop in the monitored gain is the cheap signal that something changed upstream.

Backup: three papers behind the methodological choices

Paper	Finding	What I used
Rasp & Lerch (2018), neural networks for post-processing	537 German stations, t2m: CRPS 0.78 vs 1.16 raw (≈33 % better), with station embeddings	A learned correction; lat/lon as spatial context; early stopping
Guo et al. (2017), calibration of neural networks	Deep networks are overconfident; the simplest fix wins	Simple beats flexible on small samples
Niculescu-Mizil & Caruana (2005)	Platt vs isotonic: the best choice depends on data size (≈1,000 examples)	842 rows (AIFS, Config B) call for a simpler method

Their 33 % is a CRPS gain on a calibrated ensemble; my 44.7 % is an RMSE reduction on a raw forecast with a 5.30 °C error. The agreement is in the mechanism (station information, non-linear combination), not in the percentage.

Backup: data coverage per model and variable

Rows	msl	sp	t2m	td2m	tp06
AIFS	2,136	2,497	24,611	2,508	23,700
Aurora	1,880		21,474		
DLWP			21,474		
FengWu	1,880		21,474		
FourCastNet	1,880	2,168	21,474		
FuXi	1,880		21,474		20,932
GraphCast	1,880		21,474		20,932
Pangu6	1,880		21,474		

Report Table 3.3

- AIFS is the only model with all 5 variables; DLWP has t2m only, so no Config B.
- Surface pressure: AIFS RMSE 32.7 hPa, most likely the station–grid altitude difference, not the model.
- msl RMSE: FengWu, FuXi about 1.3 hPa; GraphCast 4.9 hPa.
- Every statistic is computed per variable: averaging °C, hPa and mm has no meaning.